

05 — Domain & Data Model Specification

Canonical Domain Objects, Relationships, State Data & Persistence Baseline

Field	Value
Document ID	CloudShield AI — 05
Version	1.0
Status	Engineering Baseline
Scope	Product-level domain and data model
Primary System of Record	Relational database for transactional/domain state
Large Artifacts	Object storage for evidence, reports and large provider artifacts
Async Processing	Queue/job system with durable job state
Primary Cloud	AWS for MVP
DAIL Position	Integrated subsystem with explicit domain boundaries and lifecycle data

Data must preserve the chain from observation to trusted state.

Document Contents

1. Purpose and Scope
2. Domain Modeling Principles
3. Bounded Contexts
4. Canonical Domain Model
5. Tenant, Organization and Cloud Scope
6. Cloud Connection Domain
7. Asset and Resource Domain
8. Configuration and Observation Domain
9. Security Finding Domain
10. Risk Domain
11. Dependency Graph Domain
12. AI Intelligence Domain
13. Remediation Candidate Domain
14. DAIL Domain
15. Verification and Promotion Domain
16. Trusted State Domain
17. Evidence and Audit Domain
18. Job and Event Domain
19. Policy and Governance Domain
20. Reporting and Experimentation Domain
21. Identifiers, Versioning and Immutability
22. Data Integrity and Invariants
23. Persistence Model
24. Data Retention and Lifecycle
25. Data Access and Authorization
26. Core Data Flows
27. MVP Data Model
28. Acceptance Criteria
29. Traceability
30. Glossary

1. Purpose and Scope

This specification defines the canonical domain model and data architecture of CloudShield AI. It establishes the entities, relationships, identifiers, lifecycle data, provenance, versioning and persistence rules required for the product to maintain a coherent representation of cloud infrastructure, security posture, AI-assisted reasoning, remediation changes and trusted state.

The model is product-level. DAIL is modeled as a bounded subsystem that operates on remediation candidates, dependencies, impact, verification and promotion data. DAIL does not own the entire CloudShield AI domain.

1.1 Scope

- Logical domain entities and value objects.
- Relationships and cardinality expectations.
- State-bearing records and versioned observations.
- Provenance and evidence linkage.
- Tenant/account/cloud boundaries.
- Transactional persistence and artifact storage.
- Data integrity and lifecycle invariants.
- MVP relational data model baseline.

1.2 Out of Scope

- Provider-specific database DDL beyond the conceptual baseline.
- Complete API schemas.
- Every AWS service/resource type.
- Physical database tuning and final index selection.
- Detailed DAIL internal implementation already covered by the DAIL subsystem specifications.

2. Domain Modeling Principles

Principle	Meaning
Canonical representation	Provider-specific objects are normalized into stable CloudShield AI concepts.
Observed $\neq$ trusted	Latest provider state is not automatically the product's trusted state.
Immutable history	Material observations, decisions and state transitions remain historically traceable.
Explicit change objects	Infrastructure changes are represented as candidates, not implicit mutations.
Provenance	Important data points identify where and when they came from.
Versioned semantics	Rules, risk calculations, candidate contents and AI interactions can be versioned.
Bounded ownership	Each domain has clear ownership and mutation responsibility.
Tenant isolation	Data access is constrained by tenant and authorized cloud scope.
Evidence linkage	Material conclusions point to supporting evidence.
Fail-closed integrity	Missing critical state/evidence cannot be silently treated as valid.

Canonical product truth is assembled from multiple data classes rather than a single mutable record.

3. Bounded Contexts

Context	Primary Entities	Ownership Responsibility
Organization & Access	Tenant, User, Role, Scope	Who can access which product data.
Cloud Connectivity	CloudConnection, CloudScope	How CloudShield AI reaches cloud environments.
Inventory	Asset, ResourceIdentifier, AssetRelationship	What infrastructure exists.
Configuration	Observation, ConfigurationSnapshot, ConfigurationDiff	What infrastructure looked like at a point in time.
Security	Rule, Finding, FindingEvidence	What security/configuration conditions exist.
Risk	RiskAssessment, RiskFactor	How findings/assets are prioritized.
Graph & Impact	DependencyEdge, ImpactAnalysis, Invalidation	What relationships and blast radius exist.
AI Intelligence	AIRequest, AIResponse, AIContext, AIRecommendation	How AI reasoning is generated and traced.
Change Management	RemediationCandidate, CandidateVersion	What infrastructure change is proposed.
DAIL	DAILIdentity, LifecycleGate, VerificationRun, Promotion	Deterministic change lifecycle and safety gates.
Trusted State	TrustedState, TrustedStateVersion	What state has earned product trust.
Evidence & Audit	EvidenceArtifact, AuditEvent, Decision	Why the system believes and records something.
Operations	Job, JobAttempt, Event	Asynchronous execution and system processing.
Policy	Policy, PolicyDecision, Approval	Governance constraints and decisions.
Reporting & Evaluation	Report, Experiment, EvaluationRun	Operational reporting and controlled experimentation.

4. Canonical Domain Model

The central relationship model is:

Tenant → CloudConnection → CloudScope → Asset → Observation/Configuration → Finding → RiskAssessment → AIRecommendation → RemediationCandidate → DAILLifecycle → Verification → GovernanceDecision → Promotion → TrustedState

4.1 Core Entity Inventory

Entity	Purpose	Lifecycle/Identity Character
Tenant	Top-level product isolation boundary.	Stable immutable identity.
CloudConnection	Configured provider connection.	Mutable status + historical validation.
CloudScope	Account/region/resource scope.	Stable scope identity.
Asset	Canonical cloud resource representation.	Stable logical identity.
Observation	Provider-reported observation.	Immutable point-in-time record.
ConfigurationSnapshot	Normalized configuration state.	Versioned immutable snapshot.
ConfigurationDiff	Difference between two snapshots.	Derived immutable comparison.
SecurityRule	Deterministic security evaluation definition.	Versioned.
Finding	Security/configuration issue.	Stateful lifecycle.
RiskAssessment	Risk result for an object/finding.	Versioned calculation.
DependencyEdge	Relationship between assets.	Versioned/observed.
ImpactAnalysis	Candidate impact result.	Immutable per analysis run.
AIRequest	Structured AI invocation.	Immutable invocation record.
AIResponse	Structured model result.	Immutable response record.
RemediationCandidate	Explicit intended infrastructure change.	Stable identity + versioned content.

DAIIdentity	Lifecycle identity associated with candidate.	Stable within candidate lifecycle.
VerificationRun	Deterministic verification execution.	Immutable execution result.
Approval/Decision	Governance decision.	Immutable decision record.
Promotion	Trusted-state transition event.	Immutable event.
TrustedState	Accepted product state.	Versioned.
EvidenceArtifact	Supporting artifact.	Immutable/content-addressed where practical.
AuditEvent	Historical product event.	Append-only.
Job	Async work unit.	Stateful execution record.

5. Tenant, Organization and Cloud Scope

5.1 Tenant

- tenant_id
- display_name
- status
- created_at
- updated_at
- configuration/policy references

DATA-TEN-001: Every tenant-owned domain object shall be attributable to exactly one tenant unless explicitly classified as global configuration.

DATA-TEN-002: Cross-tenant joins and access shall be prohibited by application authorization and persistence-layer safeguards where practical.

5.2 Cloud Scope

- provider
- account/subscription/project identifier
- region/zone scope where applicable
- resource filters
- enabled services/resource types
- scope status

CloudScope defines where a discovery or analysis operation is allowed to operate; it does not itself represent an asset.

6. Cloud Connection Domain

Field	Purpose
connection_id	Stable product identity.
tenant_id	Ownership boundary.
provider	Cloud provider identifier.
provider_account_id	External account identifier.
credential_reference	Reference to secret/identity mechanism, not raw secret.
status	CONFIGURED/VALIDATING/READY/DEGRADED/DISABLED/ERROR.
last_validated_at	Validation freshness.
scope_ids	Associated cloud scopes.
created_at/updated_at	Lifecycle metadata.

DATA-CONN-001: Raw long-lived credentials shall not be stored in the ordinary domain tables.

DATA-CONN-002: Connection validation results shall be historically attributable to a connection and execution identity.

7. Asset and Resource Domain

Asset is the canonical CloudShield AI representation of an infrastructure resource. Provider-native identifiers are retained as identifiers, while product behavior uses normalized asset identity.

Asset Attribute	Description
asset_id	Stable CloudShield AI identity.
tenant_id	Tenant boundary.
asset_type	Canonical resource type.
provider	AWS in MVP; extensible.
scope_id	Cloud/account/region scope.
display_name	Human-readable name.
provider_resource_id	Provider identifier.
status	Observed lifecycle/status.
first_seen_at	First observation.
last_seen_at	Latest observation.
last_observation_id	Latest source observation.
trusted_state_id	Relevant trusted state, if modeled.
metadata	Normalized non-sensitive metadata.

7.1 Resource Identifier

- Provider-native ID.
- ARN or equivalent canonical identifier where available.
- Resource type.
- Account/scope.
- Region.
- Optional secondary identifiers.

DATA-ASSET-001: Asset identity shall be deterministic where sufficient provider identifiers exist.

DATA-ASSET-002: Identity conflicts shall not be silently merged.

8. Configuration and Observation Domain

8.1 Observation

Field	Description
observation_id	Immutable observation identity.
asset_id	Observed asset.
source	Provider/API/collector source.
observed_at	Time represented by provider observation.
collected_at	Time CloudShield AI collected it.
collector_job_id	Discovery job provenance.
raw_artifact_ref	Optional large/raw artifact reference.
normalized_payload	Canonical normalized state.
collection_status	Complete/partial/error semantics.

8.2 ConfigurationSnapshot

A configuration snapshot is a normalized, versioned representation derived from one or more observations. Snapshots should be immutable once finalized.

Snapshot Attribute	Description
snapshot_id	Immutable snapshot identity.
asset_id	Asset.
version	Monotonic or otherwise unique snapshot version.
content_hash	Content identity/integrity.
observed_at	State timestamp.
created_at	Persistence timestamp.
source_observation_ids	Provenance links.
configuration	Normalized configuration document.

8.3 ConfigurationDiff

- from_snapshot_id
- to_snapshot_id
- changed_paths
- added_paths
- removed_paths
- classification
- created_at

DATA-CONF-001: A configuration snapshot shall identify its source observation(s).

DATA-CONF-002: A proposed candidate configuration shall not overwrite an observed snapshot.

9. Security Finding Domain

9.1 SecurityRule

Attribute	Purpose
rule_id	Stable rule identity.
rule_version	Evaluation version.
name	Human-readable name.
severity	Base severity.
condition_definition	Deterministic evaluation logic/configuration.
remediation_guidance	Default remediation guidance.
enabled	Whether rule is active.

9.2 Finding

Attribute	Purpose
finding_id	Stable finding identity.
tenant_id	Tenant boundary.
rule_id/rule_version	Evaluation provenance.
asset_ids	Affected assets.
state	OPEN/ACKNOWLEDGED/IN_REMEDIATION/VERIFIED/RESOLVED/REOPENED/SUPPRESSED.
severity	Finding severity.
risk_assessment_id	Current/associated risk result.
first_detected_at	Initial detection.
last_detected_at	Latest detection.
evidence_ids	Supporting evidence.

suppression_reason	Required when suppressed.
--------------------	---------------------------

DATA-FIND-001: A finding shall preserve the rule version that generated it.

DATA-FIND-002: Finding state transitions shall be represented as auditable events.

10. Risk Domain

RiskAssessment is a versioned interpretation of a finding/asset context. The model must preserve both the resulting score and the factors used to calculate it.

Attribute	Purpose
risk_assessment_id	Assessment identity.
subject_type/subject_id	Finding/asset/candidate subject.
score	Numeric or normalized risk score.
level	Low/Medium/High/Critical or configured scale.
score_version	Calculation version.
factors	Structured factor contributions.
confidence	Assessment confidence where modeled.
calculated_at	Calculation time.
source_ids	Input/provenance references.

DATA-RISK-001: Risk calculations shall be reproducible from stored version and material inputs where practical.

DATA-RISK-002: Risk data shall not replace the underlying finding/evidence records.

11. Dependency Graph Domain

11.1 DependencyEdge

Field	Description
edge_id	Stable relationship identity.
source_asset_id	Origin asset.
target_asset_id	Related asset.
relationship_type	Canonical relationship.
direction	Directed relationship.
confidence	Relationship confidence where needed.
observed_at	Observation time.
source_evidence_ids	Evidence supporting relationship.
status	Active/invalidated/unknown.

11.2 ImpactAnalysis

Field	Description
impact_analysis_id	Analysis identity.
candidate_id	Candidate being analyzed.
graph_version/context	Graph basis.
direct_assets	Direct targets.
transitive_assets	Dependent/transitive impact.
invalidated_findings	Potentially invalidated findings.
invalidated_assumptions	Verification/trusted assumptions affected.
policy_result	Pass/block/escalate.

created_at	Analysis time.
------------	----------------

DATA-DEP-001: Dependency relationships used for safety decisions shall be attributable to evidence or defined deterministic sources.

DATA-IMP-001: Impact analysis shall record the dependency context used to derive its result.

12. AI Intelligence Domain

AI data is treated as advisory, structured and auditable. The model must support multiple providers through a stable product-level abstraction.

Entity	Key Data
AIContext	context_id, purpose, referenced object IDs, context version, authorization scope, created_at
AIRequest	request_id, provider adapter, model, prompt/template version, context_id, request metadata, timestamp
AIResponse	response_id, request_id, schema version, structured output, validation result, timestamp
AIRecommendation	recommendation_id, response_id, type, rationale, proposed actions, uncertainty/assumptions
AIProviderConfig	provider identity, model configuration reference, availability/status; secrets externalized

DATA-AI-001: AI outputs used downstream shall have a schema version and validation result.

DATA-AI-002: AI recommendations shall reference the context and source objects used to generate them.

DATA-AI-003: AI records shall not contain unrestricted production credentials.

13. Remediation Candidate Domain

The candidate is the central data object connecting recommendation to controlled infrastructure change.

Candidate Attribute	Description
candidate_id	Stable immutable candidate identity.
tenant_id	Tenant boundary.
origin_type	AI/HUMAN/RULE/TEMPLATE/AUTOMATION.
origin_reference	Source recommendation/finding/user action.
state	Lifecycle state.
version	Candidate content version.
content_hash	Exact content identity.
targets	Target asset IDs/resource identifiers.
proposed_change	Structured intended change.
expected_state	Expected post-change conditions.
policy_context	Policies relevant to candidate.
dail_identity_id	Associated DAIL lifecycle identity.
current_impact_analysis_id	Latest impact analysis.
current_verification_id	Latest verification.
created_by	Actor/service identity.
created_at/updated_at	Lifecycle metadata.

DATA-CAND-001: Candidate content must be immutable once entered into verification.

DATA-CAND-002: Candidate versions shall allow the exact verified object to be identified.

DATA-CAND-003: Candidate targets and expected state are mandatory for infrastructure-affecting candidates.

14. DAIL Domain

DAIL domain data is integrated with CloudShield AI through explicit references. DAIL controls the deterministic lifecycle of a candidate, not the broader product domain.

DAIL Entity	Purpose
DAILIdentity	Stable lifecycle identity associated with a candidate.
LifecycleGate	Identity/dependency/impact/verification/promotion gate and outcome.
DependencyAssessment	Deterministic dependency result for candidate.
InvalidationRecord	Record of state/evidence invalidated by a change.
VerificationRun	Execution and result of deterministic checks.
Promotion	Successful trusted-state transition event.
DAIEvidenceLink	Reference between DAIL events and product evidence.

DATA-DAIL-001: DAIL records shall reference the CloudShield AI candidate they govern.

DATA-DAIL-002: DAIL gate results shall be attributable to an exact candidate version.

DATA-DAIL-003: DAIL promotion records shall reference the resulting trusted-state version.

15. Verification and Promotion Domain

15.1 VerificationRun

Field	Purpose
verification_id	Verification execution identity.
candidate_id/version	Exact candidate under test.
checks	Structured verification checks.
expected_state	Expected outcome basis.
observed_state	Actual observed result.
status	RUNNING/PASSED/FAILED/PARTIAL.
failure_details	Structured failure reason.
evidence_ids	Verification evidence.
executed_at	Execution timestamp.

15.2 Promotion

Field	Purpose
promotion_id	Promotion identity.
candidate_id/version	Promoted candidate.
verification_id	Required successful verification.
decision_id	Required governance decision where applicable.
from_trusted_state_id	Previous trusted state.
to_trusted_state_id	New trusted state.
status	PROMOTING/SUCCEEDED/FAILED.
promoted_at	Completion time.

DATA-PROM-001: A successful promotion shall reference successful verification and all required governance decisions.

DATA-PROM-002: A failed promotion shall not create a new trusted state.

16. Trusted State Domain

TrustedState represents the product's accepted infrastructure state for a defined asset/scope. It is separate from the latest observed configuration.

Field	Purpose
trusted_state_id	Stable trusted-state identity.
asset_id/scope_id	Subject scope.
version	Trusted-state version.
configuration_snapshot_id	Accepted configuration basis.
source_promotion_id	Promotion that established the state.
status	ACTIVE/INVALIDATED/SUPERSEDED.
created_at	Trust establishment time.
invalidated_at	If invalidated.

DATA-TRUST-001: Trusted state shall only be created or advanced by a successful governed promotion path.

DATA-TRUST-002: Trusted state invalidation shall be recorded, not silently overwritten.

DATA-TRUST-003: Observed configuration must remain independently queryable after promotion.

17. Evidence and Audit Domain

17.1 EvidenceArtifact

Field	Purpose
evidence_id	Evidence identity.
tenant_id	Tenant boundary.
type	Observation/analysis/AI/verification/decision/operational.
source_type/source_id	Originating object.
content_ref	Reference to object storage or structured content.
content_hash	Integrity/content identity.
captured_at	Evidence time.
retention_class	Retention policy.
classification	Sensitivity/access classification.

17.2 AuditEvent

- event_id
- tenant_id
- actor_type
- actor_id
- event_type
- object_type/object_id
- previous_state
- new_state

- reason
- correlation_id
- created_at

DATA-EVID-001: Material product state transitions shall generate auditable records.

DATA-EVID-002: Evidence content shall be integrity-protected through content hashing or equivalent mechanisms where practical.

DATA-EVID-003: Audit records shall be append-only from the application perspective.

18. Job and Event Domain

Job Field	Purpose
job_id	Stable async operation identity.
job_type	DISCOVERY/ANALYSIS/AI/VERIFICATION/REPORT/etc.
tenant_id	Tenant scope.
scope	Execution scope.
status	QUEUED/RUNNING/SUCCEEDED/PARTIAL/FAILED/CANCELLED.
attempt_count	Retry count.
correlation_id	Cross-service trace.
input_reference	Input object/job request.
output_reference	Produced result.
error	Structured failure details.
timestamps	Lifecycle timing.

18.1 Event

- event_id
- event_type
- aggregate_type/aggregate_id
- tenant_id
- payload_version
- occurred_at
- correlation_id
- causation_id

DATA-JOB-001: Jobs shall be durable enough for users and operators to inspect their final outcome.

DATA-JOB-002: Events shall be versioned when consumed asynchronously.

DATA-JOB-003: Irreversible transitions shall be idempotency-aware.

19. Policy and Governance Domain

Entity	Purpose
Policy	Defines technical/governance constraints.
PolicyVersion	Immutable policy definition version.
PolicyEvaluation	Result of applying policy to a candidate/operation.
ApprovalRequest	Decision request requiring human or configured approval.
Decision	APPROVE/REJECT/REQUEST_CHANGES/ESCALATE/AUTO_APPROVE outcome.
DecisionReason	Required rationale where policy demands it.

DATA-POL-001: Governance decisions shall reference the policy context applicable at decision time.

DATA-POL-002: Approval/rejection decisions shall be immutable historical records.

20. Reporting and Experimentation Domain

Entity	Core Fields
Report	report_id, tenant_id, type, scope, generated_at, data_version, artifact_ref
Experiment	experiment_id, configuration, hypothesis, owner, status, created_at
EvaluationRun	evaluation_id, experiment_id, dataset/context, metrics, results, timestamp
BenchmarkResult	metric, score, version, environment, evidence reference

Experimental records must be clearly separated from production trusted-state records. An experiment cannot silently promote its result into production state.

21. Identifiers, Versioning and Immutability

21.1 Identifier Rules

- Use globally unique identifiers for major domain entities.
- Provider identifiers remain available for reconciliation and traceability.
- Human-readable names are not identity.
- Candidate content identity should use version and/or content hash.
- Evidence artifacts should use content hashes where practical.
- Correlation IDs connect related asynchronous actions.

21.2 Versioning Rules

Object	Versioning Requirement
Observation	Immutable by observation ID.
ConfigurationSnapshot	Immutable snapshot/version.
SecurityRule	Explicit rule version.
RiskAssessment	Explicit calculation version.
AI contract	Schema/template/model metadata.
Candidate	Immutable verification version; new content creates new version.
Policy	Explicit policy version.
Verification	Immutable execution record.
TrustedState	Monotonic history of accepted versions.
Evidence/Audit	Append-only history.

21.3 Immutability Boundary

The product may update operational metadata such as status timestamps, but must not mutate historical facts that are needed to explain why a decision or trusted-state transition occurred.

22. Data Integrity and Invariants

INV-01: Every tenant-owned object has a valid tenant boundary.

INV-02: Every observation references a valid asset and source execution context.

INV-03: Every finding references the rule version that generated it.

INV-04: Every risk assessment identifies its calculation version.

INV-05: Every infrastructure-affecting candidate identifies exact targets and expected state.

INV-06: A candidate under verification cannot be modified in place.

INV-07: Every DAIL gate result references an exact candidate version.

INV-08: Promotion cannot succeed without required verification success.

INV-09: Trusted state cannot advance on promotion failure.

INV-10: Observed state remains independently stored from trusted state.

INV-11: Material state transitions generate audit/evidence records.

INV-12: AI recommendations remain attributable to a request, response and context.

INV-13: Suppressed findings retain suppression reason and historical evidence.

INV-14: Partial/unknown collection results cannot be represented as complete observations.

INV-15: Experiment data cannot directly mutate trusted production state.

23. Persistence Model

23.1 Relational System of Record

The MVP should use a relational database as the primary system of record for transactional entities, relationships, state transitions and metadata. This provides strong consistency for lifecycle operations and supports relational queries across assets, findings, candidates, evidence references and governance.

23.2 Object Storage

- Large raw provider responses.
- Configuration/export artifacts.
- Verification artifacts.
- Generated reports.
- Large AI/evaluation artifacts where retention policy allows.
- Archived evidence.

23.3 Queue/Job Store

Asynchronous work should use durable queueing plus job state persisted in the system of record. The queue is not the source of truth for final lifecycle state.

23.4 Optional Graph Layer

A dedicated graph database is not required for the MVP. Dependency relationships can initially be modeled relationally. A graph-optimized store can be introduced if scale/query characteristics justify it, without changing the canonical domain model.

24. Data Retention and Lifecycle

Data	Default Lifecycle Principle
Current assets	Retain while within tenant/cloud scope; historical status retained.
Observations	Retain according to evidence/operational retention policy.
Configuration snapshots	Retain enough history for drift, audit and troubleshooting.
Findings	Retain lifecycle history; resolved records remain traceable.
AI records	Retain according to privacy, security and evaluation policy.
Candidates	Retain complete lifecycle history.
Verification	Retain as decision evidence.
Trusted states	Retain all historical accepted versions.
Audit events	Long-lived append-only retention subject to policy.
Raw artifacts	Use explicit retention class and lifecycle policy.

DATA-RET-001: Retention policies shall not delete records required to explain a material security or trusted-state decision within the defined audit window.

DATA-RET-002: Deletion/expiry operations shall themselves be auditable where required.

25. Data Access and Authorization

- Tenant boundary is the first authorization filter.
- Cloud account/scope is a second authorization boundary.
- Role determines allowed operations.
- Sensitive evidence may require additional classification-based access.
- Workers receive only data needed for their operation.
- AI context assembly must enforce the same authorization boundaries as ordinary product reads.
- Audit records must not become a side channel for secrets or unauthorized data.

DATA-AUTH-001: An AI request shall not receive context that the initiating actor/service could not otherwise access.

DATA-AUTH-002: Cross-tenant data must be inaccessible through ordinary identifiers or query manipulation.

DATA-AUTH-003: Sensitive evidence access shall be authorization-checked.

26. Core Data Flows

26.1 Discovery Flow

31. Job created.
32. Cloud connection and scope validated.
33. Provider observations collected.
34. Raw artifacts stored when required.
35. Observations persisted.
36. Assets created/reconciled.
37. Configuration snapshots derived.
38. Analysis jobs created.
39. Evidence links recorded.

26.2 Finding-to-Candidate Flow

40. Finding references asset and evidence.
41. AI context references finding, asset and permitted dependencies.
42. AI response produces recommendation.
43. Recommendation becomes candidate content.
44. Candidate receives identity/version.
45. DAIL lifecycle records gate data.
46. Impact and verification reference exact candidate version.

26.3 Promotion-to-Trusted-State Flow

47. Verification succeeds.
48. Governance decision succeeds.
49. Promotion record references candidate, verification and decision.
50. New configuration/trusted state version is created.
51. Promotion event is recorded.
52. Observed state is refreshed.
53. Finding/risk state is recalculated.
54. Evidence chain is complete.

27. MVP Data Model

The MVP should implement the following minimum entities before broadening resource/service coverage:

Priority	Entities
P0	Tenant, User/Role, CloudConnection, CloudScope, Asset, ResourceIdentifier
P0	Observation, ConfigurationSnapshot, ConfigurationDiff
P0	SecurityRule, Finding, FindingEvidence, RiskAssessment
P0	DependencyEdge, ImpactAnalysis
P0	AIContext, AIRequest, AIResponse, AIRecommendation
P0	RemediationCandidate, CandidateVersion
P0	DAILIdentity, LifecycleGate, VerificationRun, Promotion
P0	TrustedState, EvidenceArtifact, AuditEvent
P0	Job, Event, Policy, ApprovalRequest, Decision
P1	Report, Experiment, EvaluationRun, advanced graph structures

The MVP data model should prove the complete control loop before introducing additional storage technologies or large-scale specialization.

28. Acceptance Criteria

DATA-ACC-01: Every major product entity has a stable identity and tenant boundary.

DATA-ACC-02: Cloud assets can be reconciled from repeated observations.

DATA-ACC-03: Observations and configuration snapshots preserve provenance and timestamps.

DATA-ACC-04: Observed configuration can be compared with historical configuration.

DATA-ACC-05: Findings preserve rule/version/evidence provenance.

- DATA-ACC-06:** Risk assessments preserve score version and major contributing factors.
- DATA-ACC-07:** Dependency relationships are represented independently of UI graph rendering.
- DATA-ACC-08:** Impact analyses preserve the graph context used for their result.
- DATA-ACC-09:** AI interactions preserve context, request, response, validation and recommendation relationships.
- DATA-ACC-10:** Infrastructure changes are represented as versioned remediation candidates.
- DATA-ACC-11:** DAIL records reference exact candidate versions.
- DATA-ACC-12:** Verification results are immutable and evidence-linked.
- DATA-ACC-13:** Governance decisions are immutable and attributable.
- DATA-ACC-14:** Promotion records connect candidate, verification, decision and trusted state.
- DATA-ACC-15:** Trusted state is separate from observed state.
- DATA-ACC-16:** Drift/invalidation can be represented without overwriting history.
- DATA-ACC-17:** Audit/evidence records provide an explainable chain across material transitions.
- DATA-ACC-18:** Async jobs have durable state and correlation identifiers.
- DATA-ACC-19:** Data access can be constrained by tenant, cloud scope and role.
- DATA-ACC-20:** The relational MVP model can evolve toward specialized graph/object storage without changing canonical domain semantics.

29. Traceability

Domain/Data Area	Functional Specification Alignment	UX Alignment
Cloud connection	Cloud onboarding	Connection setup/status
Asset/observation	Discovery/inventory	Asset list/detail
Configuration	Snapshots/drift	Configuration diff
Finding	Security analysis	Findings experience
Risk	Risk prioritization	Risk detail
Dependency/impact	Dependency/impact workflow	Impact view
AI	AI intelligence workflow	AI insight panel
Candidate	Remediation lifecycle	Candidate detail
DAIL	DAIL integration	DAIL lifecycle view
Verification/promotion	Governance and promotion	Approval/promotion UX
Trusted state	Trusted state/monitoring	Trusted/drift views
Evidence	Evidence/audit	Evidence timeline
Jobs/events	Async processing	Job center/notifications

This document provides the canonical data vocabulary that subsequent lifecycle, identity/dependency, security, AI and infrastructure specifications must use unless a deliberate architectural decision supersedes it.

30. Glossary

Term	Definition
Domain Entity	Persistent concept with identity and lifecycle within CloudShield AI.
Observation	Provider-reported state captured at a point in time.
Configuration Snapshot	Normalized version of an asset's observed configuration.
Finding	Security/configuration condition identified by analysis.
Risk Assessment	Versioned prioritization result and contributing factors.
Dependency Edge	Relationship between infrastructure assets.
Impact Analysis	Deterministic assessment of resources/assumptions affected by a candidate.
AI Context	Authorized structured context supplied to an AI invocation.
Candidate	Explicit intended infrastructure change.
DAIL Identity	Lifecycle identity attached to a remediation candidate.
Verification	Deterministic evaluation of expected post-change state.
Promotion	Governed transition establishing a new trusted state.
Trusted State	Product state accepted after required gates.
Evidence Artifact	Artifact supporting an observation, analysis, decision or transition.
Audit Event	Historical record of a material system/action transition.
Correlation ID	Identifier linking related asynchronous operations and events.

Document Completion Statement

CloudShield AI Domain & Data Model Specification v1.0 establishes the canonical data vocabulary and persistence baseline for the product. It deliberately separates observed cloud reality, analytical conclusions, AI-generated recommendations, explicit change candidates, deterministic verification and trusted state.

The next document is 06 — CloudShield AI State & Lifecycle Specification, which will formalize the state machines, transition rules and lifecycle invariants that operate over this domain model.